

Physical Chemistry of Materials

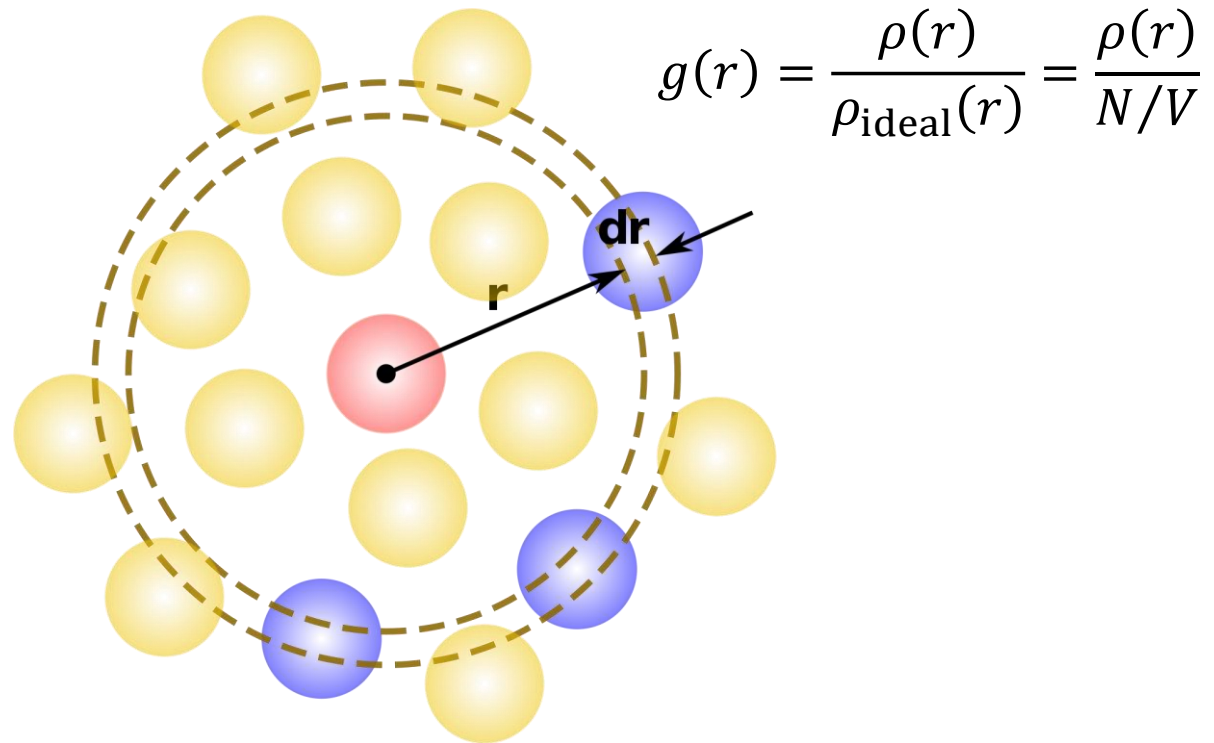
Lecture 19

Single-Chain Structures of Polymers

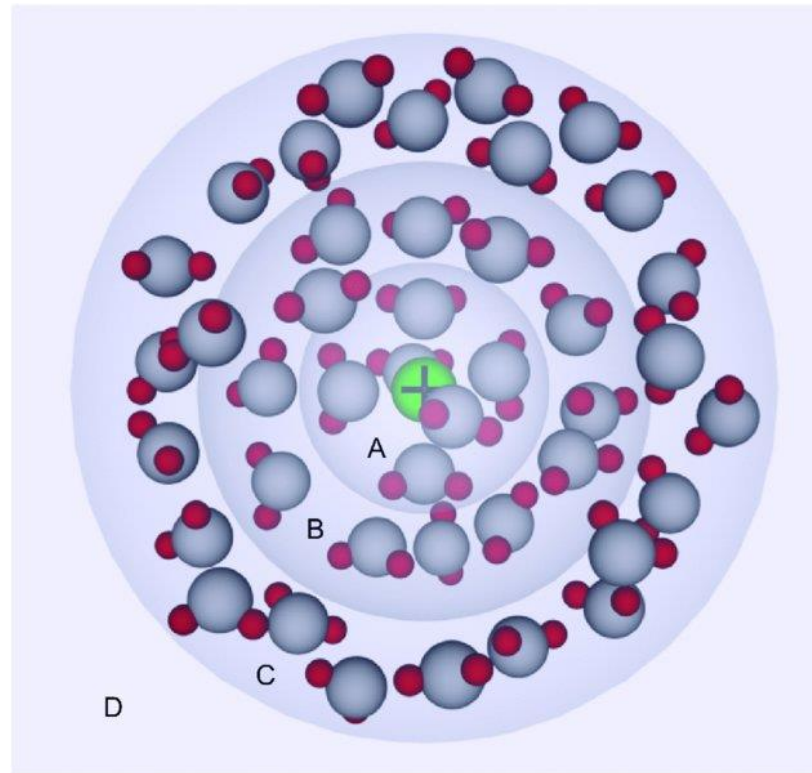


Last Class

Radial Distribution Function

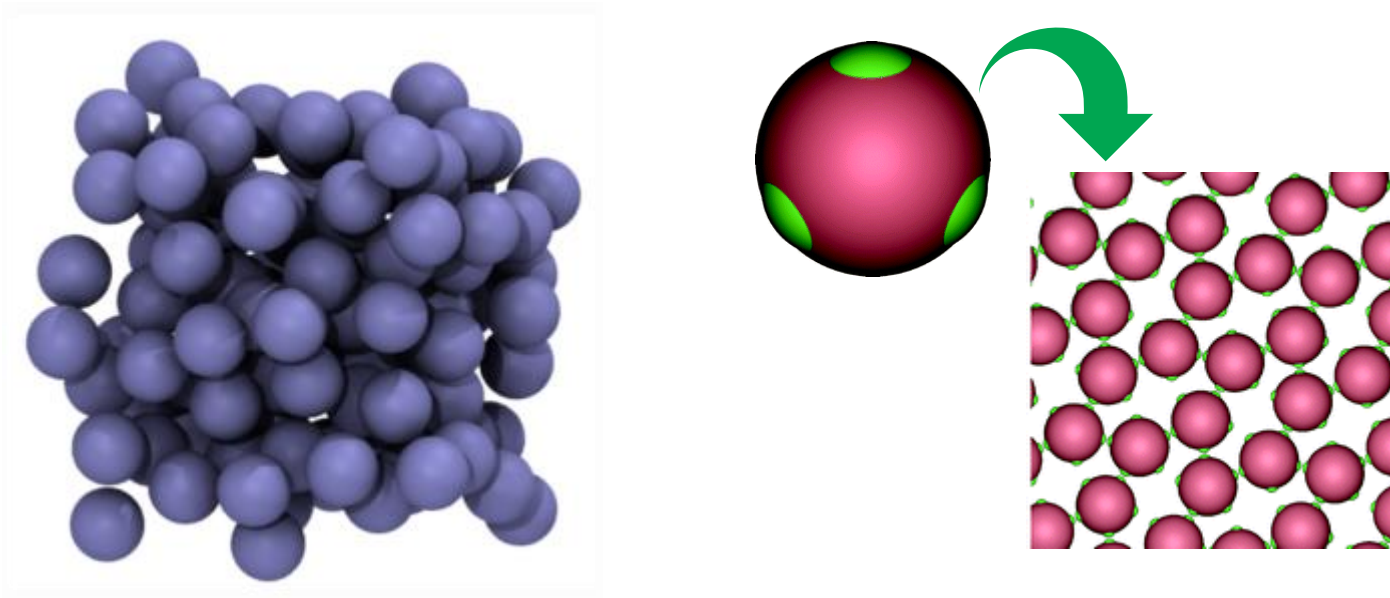


Solvation Shells



(Image: <https://basicmedicalkey.com/solutions-of-electrolytes/>)

Simple vs. Network Fluids



(Images: <https://www.pyretis.org/current/user/getting-started.html>
Bianchi et al., *Phys Chem Chem Phys* (2011))

Three Moduli

- Bulk modulus K (all phases)

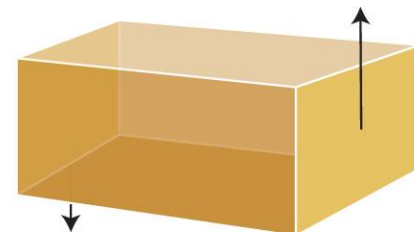
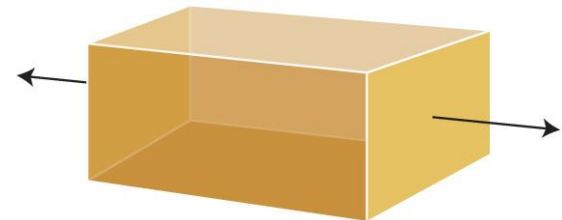
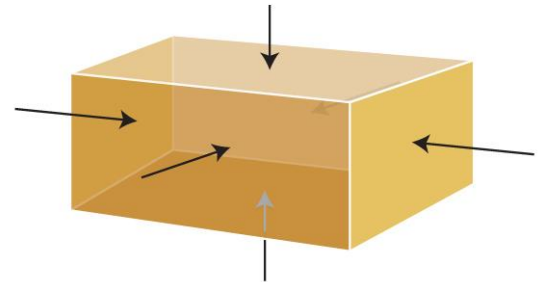
$$K = \frac{\text{pressure}}{\text{relative volume change}}$$

- Young's modulus E (solids)

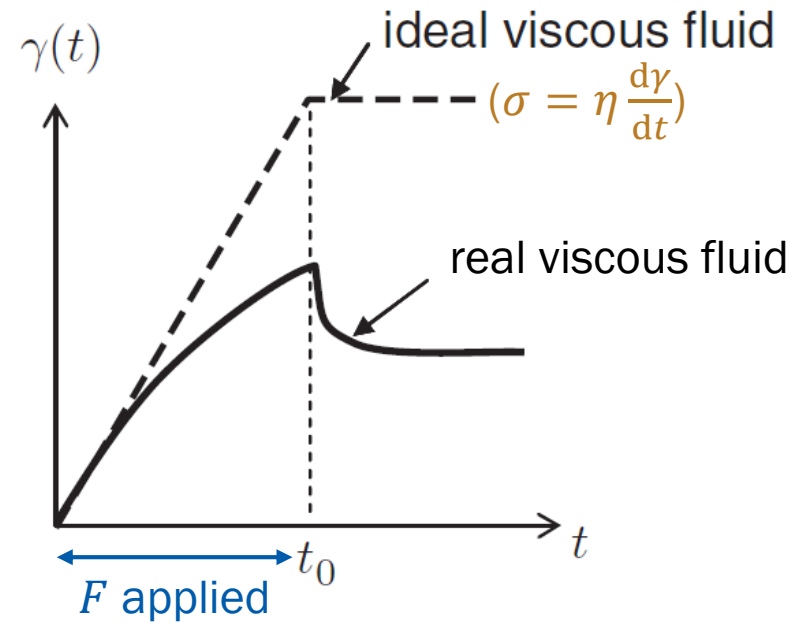
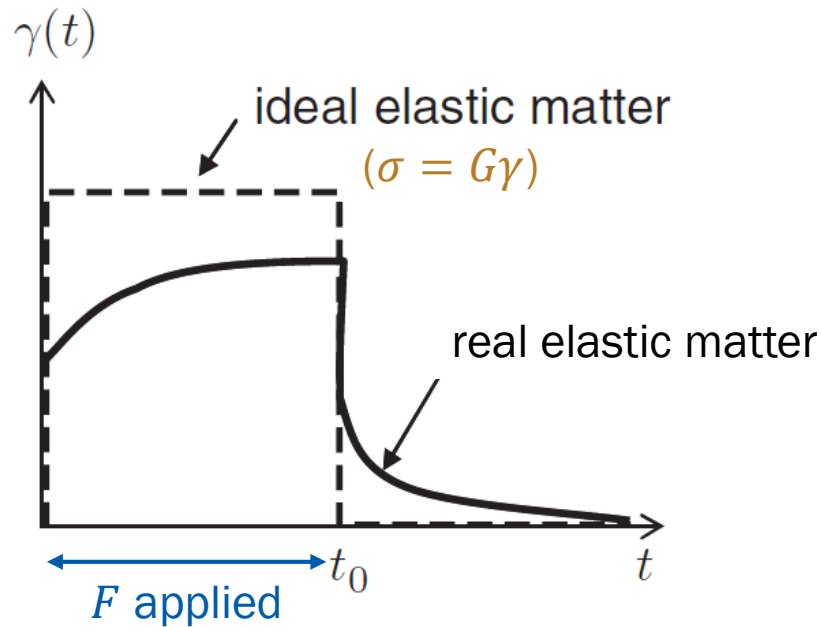
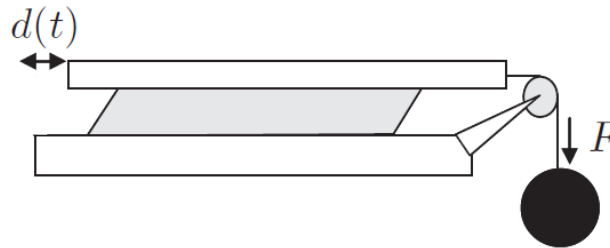
$$E = \frac{\text{uniaxial stress}}{\text{uniaxial strain}}$$

- Shear modulus G (solids)

$$G = \frac{\text{shear stress}}{\text{shear strain}}$$

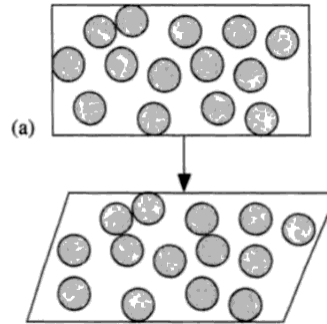


Elasticity and Viscosity

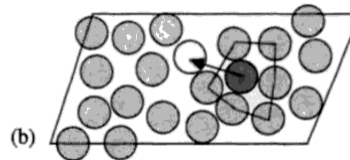


Molecular View on Viscosity

- Instantaneous response of a liquid: no rearrangement

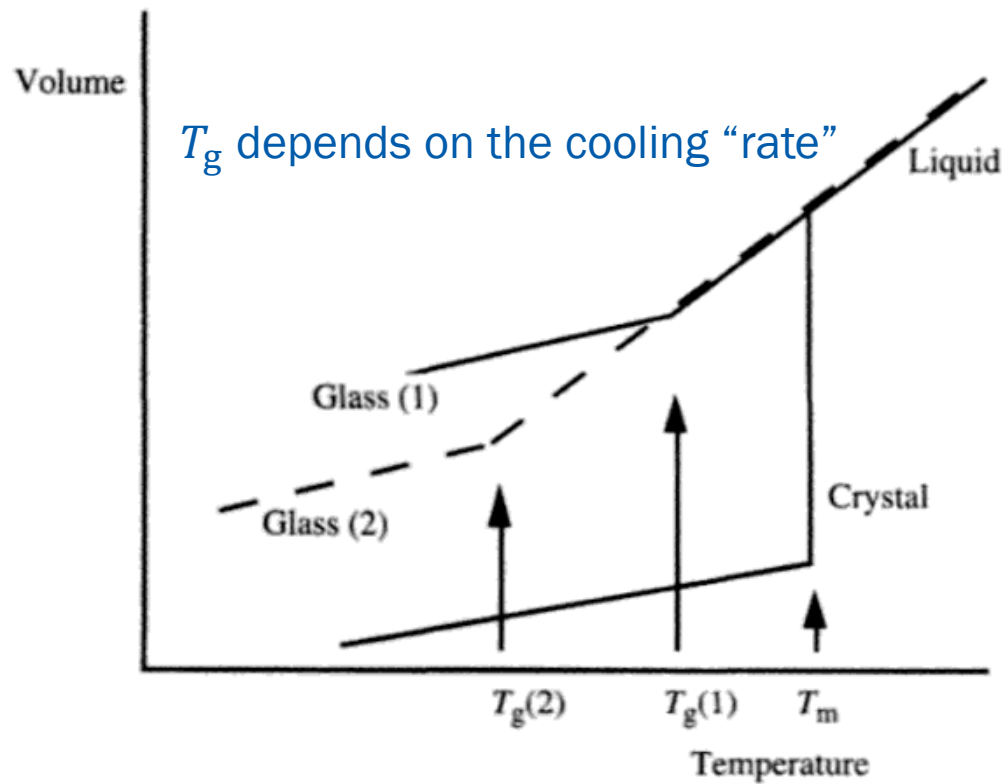


- Relaxation: escape from a “cage” by neighbors



Glasses

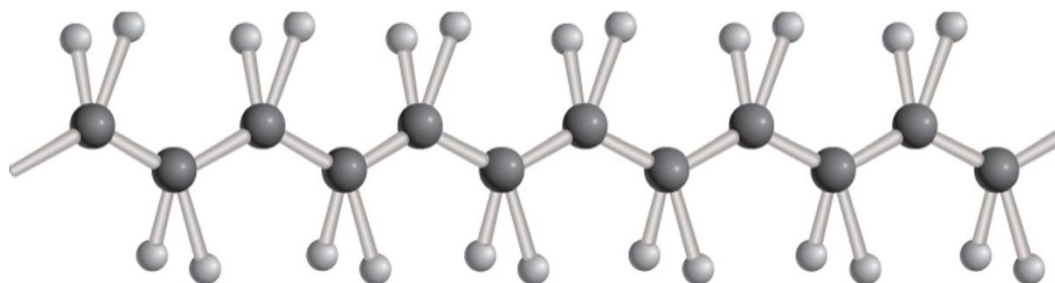
$$\tau \gtrsim \tau_{\text{exp}}$$



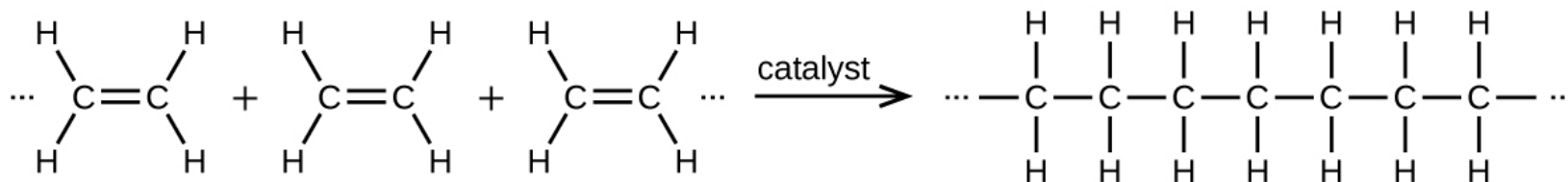
(Image: Jones, *Soft Condensed Matter*, Fig. 2.10)

Polymers

Large molecules made of multiple simpler chemical units



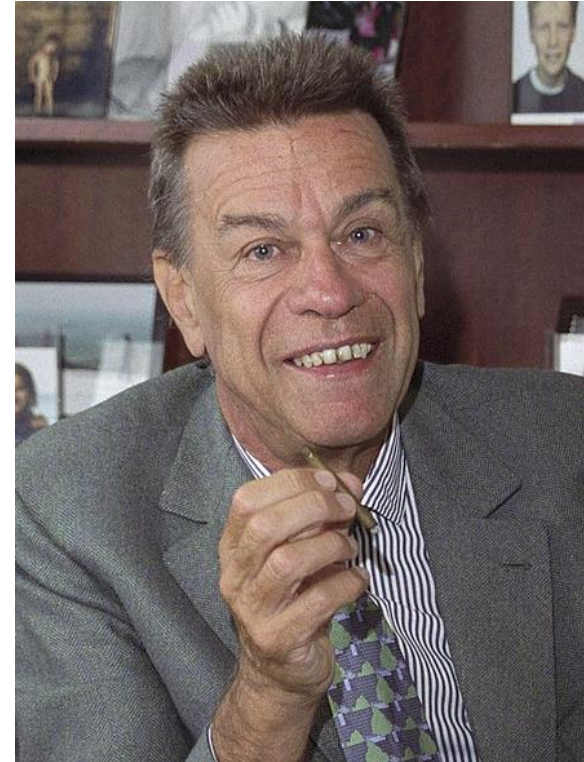
- **Monomers:** repeating units



Nobel Laureates: Polymer Theory



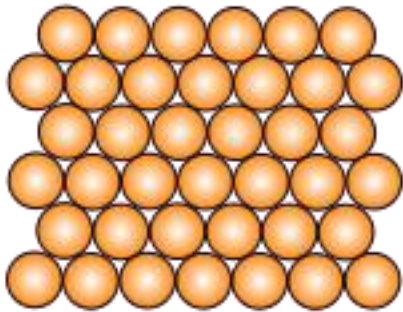
Paul Flory
(Chemistry 1974)



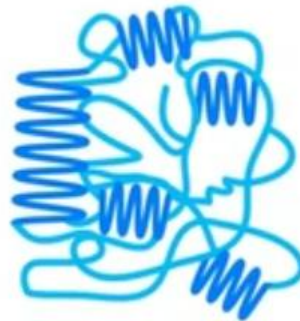
Pierre-Gilles de Gennes
(Physics 1991)

Polymer Melts

crystalline solid

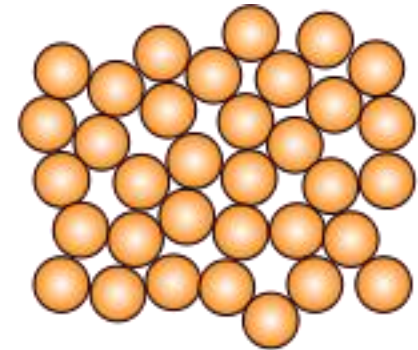


semi-crystalline solid

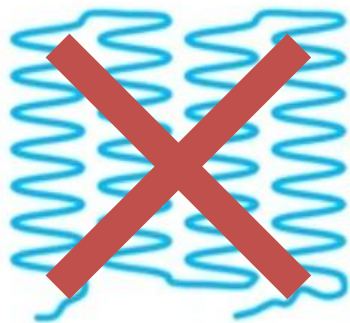


rigid, high-density

amorphous solid



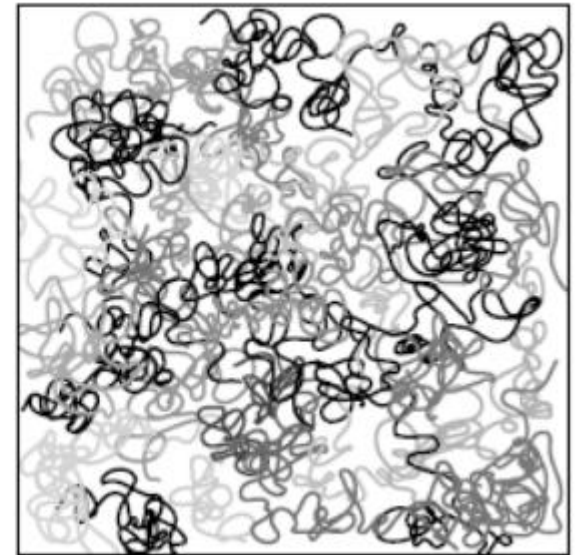
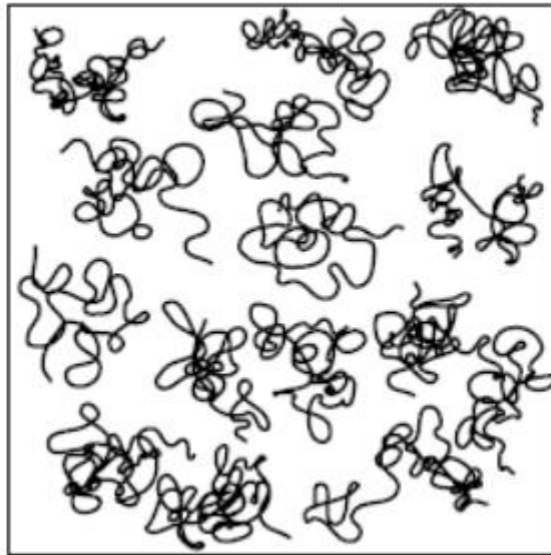
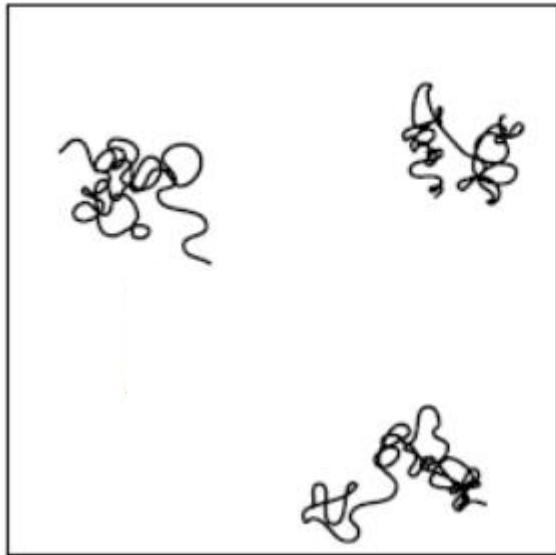
soft, low-density



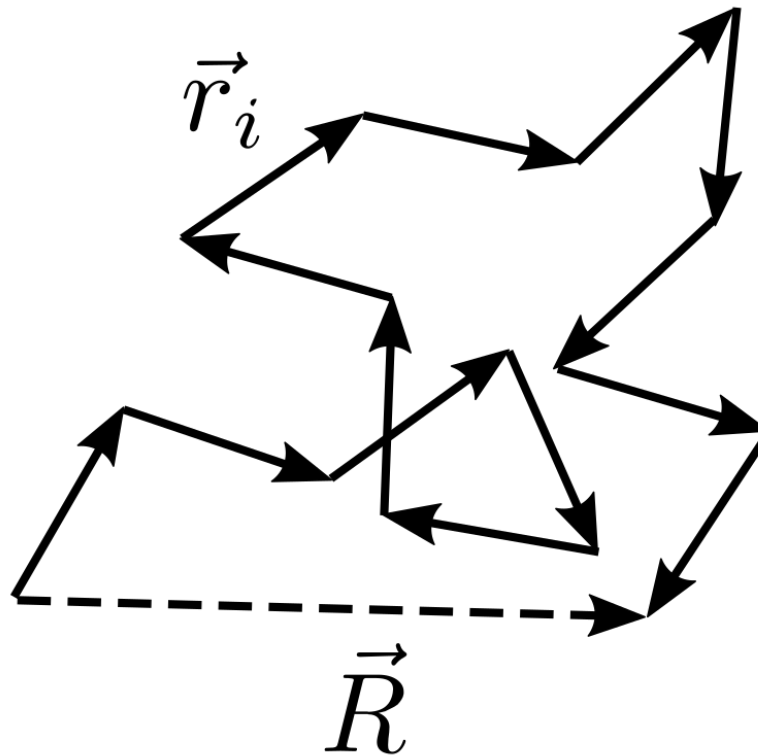
(Image: https://commons.wikimedia.org/wiki/File:Crystalline_polycrystalline_amorphous2.svg)

Polymer Solutions

polymer concentration



Ideal Chains



(Image: https://commons.wikimedia.org/wiki/File:Ideal_chain_random_walk.svg)

Freely Jointed Chain (FJC)

For the chain of N links (size a), the end-to-end vector $\vec{R}$ is

$$\vec{R} = \sum_{i=1}^N \vec{r}_i$$

If each link is unbiased,

$$\langle \vec{R} \rangle = \sum_i \langle \vec{r}_i \rangle = 0$$

$$\langle \vec{R}^2 \rangle = \left\langle \sum_{i,j} \vec{r}_i \cdot \vec{r}_j \right\rangle = Na^2$$

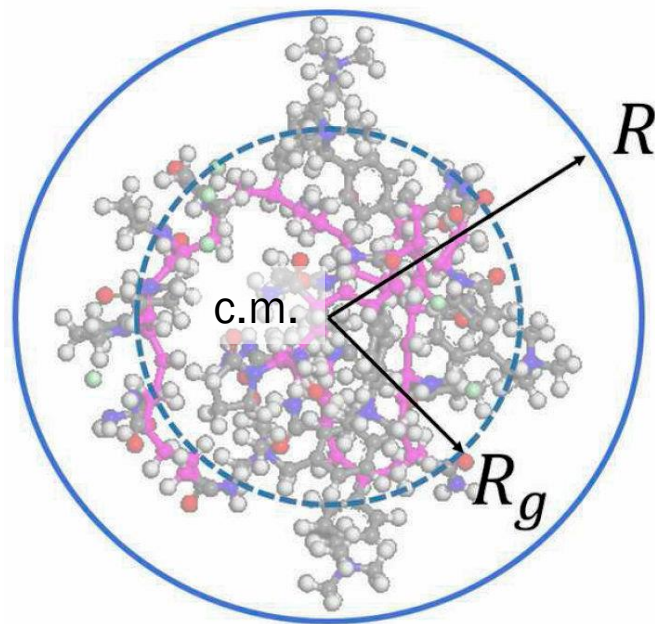
Freely Jointed Chain

$$\langle \vec{R}^2 \rangle = \left\langle \sum_{i,j} \vec{r}_i \cdot \vec{r}_j \right\rangle = Na^2$$

Hence, the “overall size” of the chain can be quantified as

$$\sqrt{\langle \vec{R}^2 \rangle} = \sqrt{Na} \propto \sqrt{N}$$

Radius of Gyration R_g



$$R_g^2 = \frac{1}{N} \sum_i^N (\vec{r}_i - \vec{r}_{\text{cm}})^2$$

(Image: <http://dx.doi.org/10.31349/RevMexFis.65.321>)

Radius of Gyration R_g

$$R_g^2 = \frac{1}{N} \sum_i^N (\vec{r}_i - \vec{r}_{\text{cm}})^2$$

As $\vec{r}_{\text{cm}} = \frac{1}{N} \sum_i^N \vec{r}_i$, one can show that

$$R_g^2 = \frac{1}{N^2} \sum_{i,j}^{N,N} (\vec{r}_i^2 - \vec{r}_i \cdot \vec{r}_j) = \frac{1}{2N^2} \sum_{i,j}^{N,N} (\vec{r}_i - \vec{r}_j)^2$$

$$\langle R_g^2 \rangle = \frac{1}{2N^2} \sum_{i,j}^{N,N} \langle (\vec{r}_i - \vec{r}_j)^2 \rangle$$

R_g of FJC

$$\langle R_g^2 \rangle = \frac{1}{2N^2} \sum_{i,j}^{N,N} \langle (\vec{r}_i - \vec{r}_j)^2 \rangle = \frac{1}{N^2} \sum_{i=1}^N \sum_{j=i}^N \langle (\vec{r}_i - \vec{r}_j)^2 \rangle$$

Using the end-to-end distance $\langle (\vec{r}_i - \vec{r}_j)^2 \rangle = (j - i)a^2$,

$$\sum_{i=1}^N \sum_{j=i}^N (j - i) \approx \int_0^N \int_u^N (v - u) dv du = \frac{N^3}{6}$$

$$\langle R_g^2 \rangle = \frac{Na^2}{6} \propto N$$

Power Laws

For a freely jointed chain,

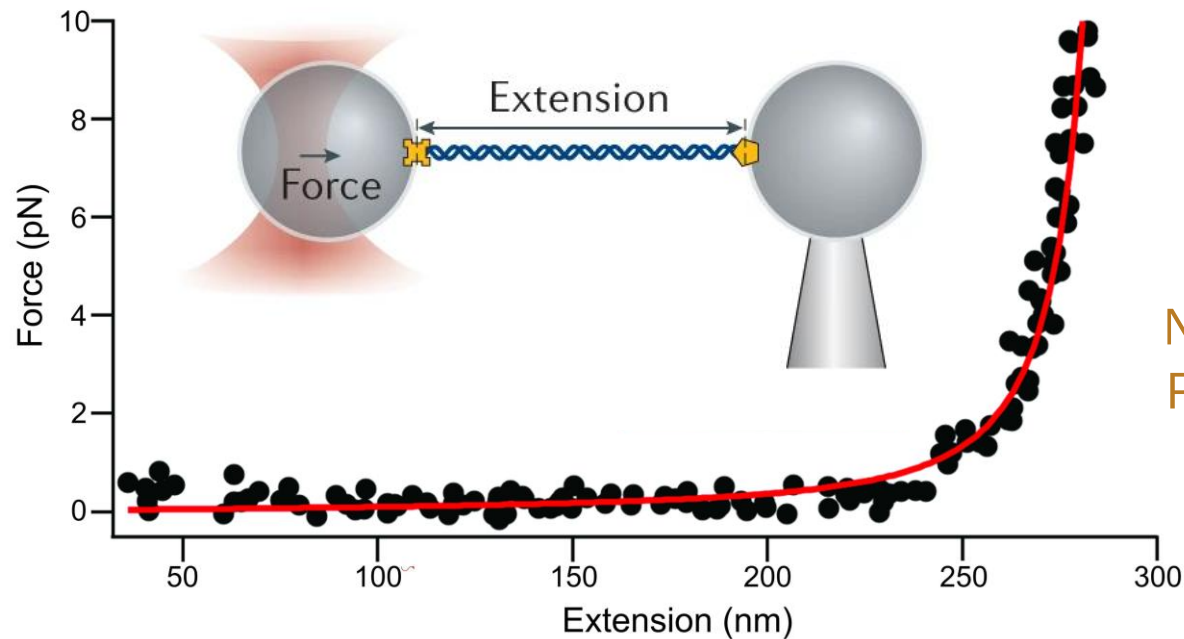
- End-to-end distance

$$\langle \vec{R}^2 \rangle^{1/2} = (Na^2)^{1/2} \propto N^{1/2}$$

- Radius of gyration

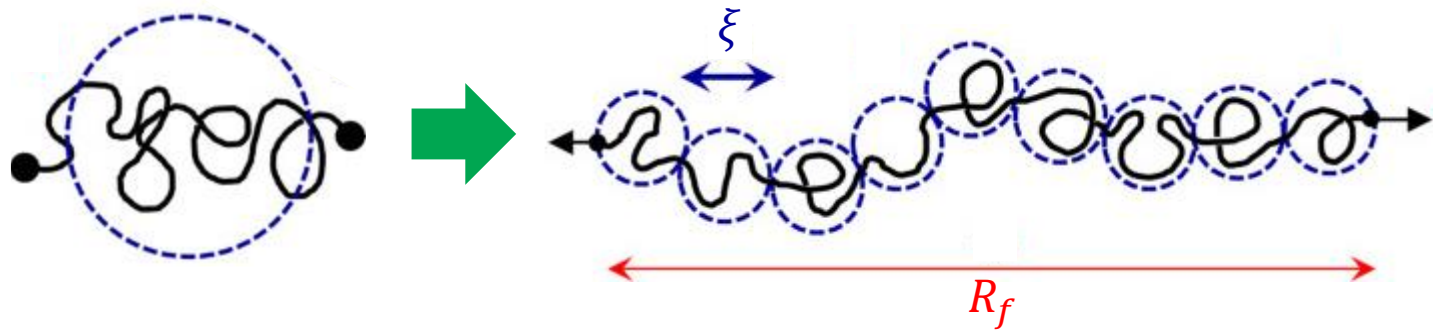
$$\langle R_g^2 \rangle^{1/2} = \left(\frac{Na^2}{6} \right)^{1/2} \propto N^{1/2}$$

Force-Extension Experiment



Nobel prize in
Physics 2018

Chain Stretching: Blob Argument



Inside a blob (of g monomers), ideal chain statistics holds:

$$\xi^2 = a^2 g$$

$$R_f \approx \frac{\xi N}{g} = \frac{Na}{\sqrt{g}}$$

Thermodynamics of Stretching

$$R_f \approx \frac{\xi N}{g} = \frac{Na}{\sqrt{g}}$$

- Energy $U = 0$ (ideal chain)
- Entropy S : restriction of a degree of freedom per blob

$$S \approx -k_B \frac{N}{g} = -k_B \frac{R_f^2}{Na^2}$$

- Free energy $A = U - TS = k_B T \frac{R_f^2}{Na^2}$

More Formally

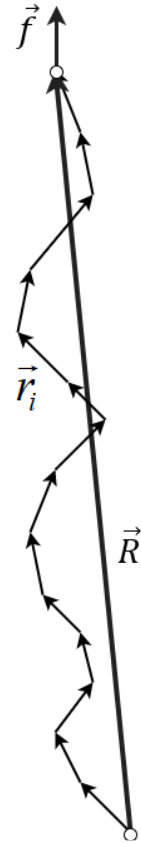
For applied force $\vec{f}$, system energy is given by

$$E = -\vec{f} \cdot \vec{R} = -\vec{f} \cdot \sum_{i=1}^N \vec{r}_i$$

The partition function is then

$$Z = \int \cdots \int e^{-\beta E} d\Omega_1 \cdots d\Omega_N$$

where $d\Omega_i = \sin \theta_i d\theta_i d\phi_i$



Partition Function

$$Z = \int \cdots \int e^{\beta \vec{f} \cdot \sum_{i=1}^N \vec{r}_i} d\Omega_1 \cdots d\Omega_N$$

$$Z = \int \cdots \int \prod_{i=1}^N e^{\beta \vec{f} \cdot \vec{r}_i} d\Omega_1 \cdots d\Omega_N$$

$$Z = \prod_{i=1}^N \left(\int e^{\beta \vec{f} \cdot \vec{r}_i} d\Omega_i \right) = \left(\int e^{\beta \vec{f} \cdot \vec{r}} d\Omega \right)^N$$

Partition Function

$$Z = \left(\int e^{\beta \vec{f} \cdot \vec{r}} d\Omega \right)^N = z^N$$

Assuming $\vec{f} \parallel \hat{k}$ (z-axis),

$$z = \int e^{\beta \vec{f} \cdot \vec{r}} d\Omega = \int_0^{2\pi} \int_0^\pi e^{\beta f a \cos \theta} \sin \theta d\theta d\phi$$

By letting $x = \cos \theta$,

$$z = \int_{-1}^1 e^{\beta f a x} dx \int_0^{2\pi} 1 d\phi = \frac{e^{\beta f a} - e^{-\beta f a}}{\beta f a} \times 2\pi$$

Partition Function

$$z = \frac{e^{\beta f a} - e^{-\beta f a}}{\beta f a} \times 2\pi$$

$$z = \frac{4\pi k_B T}{f a} \sinh\left(\frac{f a}{k_B T}\right)$$

Hence,

$$Z = z^N = \left[\frac{4\pi k_B T}{f a} \sinh\left(\frac{f a}{k_B T}\right) \right]^N$$

Extension

Now consider $\langle \vec{R} \rangle$. By symmetry, $\vec{f} \parallel \hat{k}$ leads to $\langle \vec{R} \rangle \parallel \hat{k}$.

$$|\langle \vec{R} \rangle| = \langle \vec{R} \cdot \hat{k} \rangle = \langle \sum_{i=1}^N \vec{r}_i \cdot \hat{k} \rangle = \langle \sum_{i=1}^N a \cos \theta_i \rangle$$

However,

$$Z = \int \cdots \int e^{\beta \vec{f} \cdot \sum_{i=1}^N \vec{r}_i} d\Omega_1 \cdots d\Omega_N = \int \cdots \int e^{\beta f \sum_{i=1}^N a \cos \theta_i} d\Omega_1 \cdots d\Omega_N$$

and

$$|\langle \vec{R} \rangle| = \frac{1}{\beta} \frac{\partial \ln Z}{\partial f} = k_B T \frac{\partial \ln Z}{\partial f}$$

Extension

$$|\langle \vec{R} \rangle| = k_B T \frac{\partial \ln Z}{\partial f}$$

$$Z = \left[\frac{4\pi k_B T}{fa} \sinh \left(\frac{fa}{k_B T} \right) \right]^N$$

Combining the two, we obtain

$$|\langle \vec{R} \rangle| = N k_B T \left[-\frac{1}{f} + \frac{a}{k_B T} \coth \left(\frac{fa}{k_B T} \right) \right]$$

$$|\langle \vec{R} \rangle| = Na \left[\coth \left(\frac{fa}{k_B T} \right) - \frac{k_B T}{fa} \right]$$

Extension

$$|\langle \vec{R} \rangle| = Na \left[\coth \left(\frac{fa}{k_B T} \right) - \frac{k_B T}{fa} \right]$$

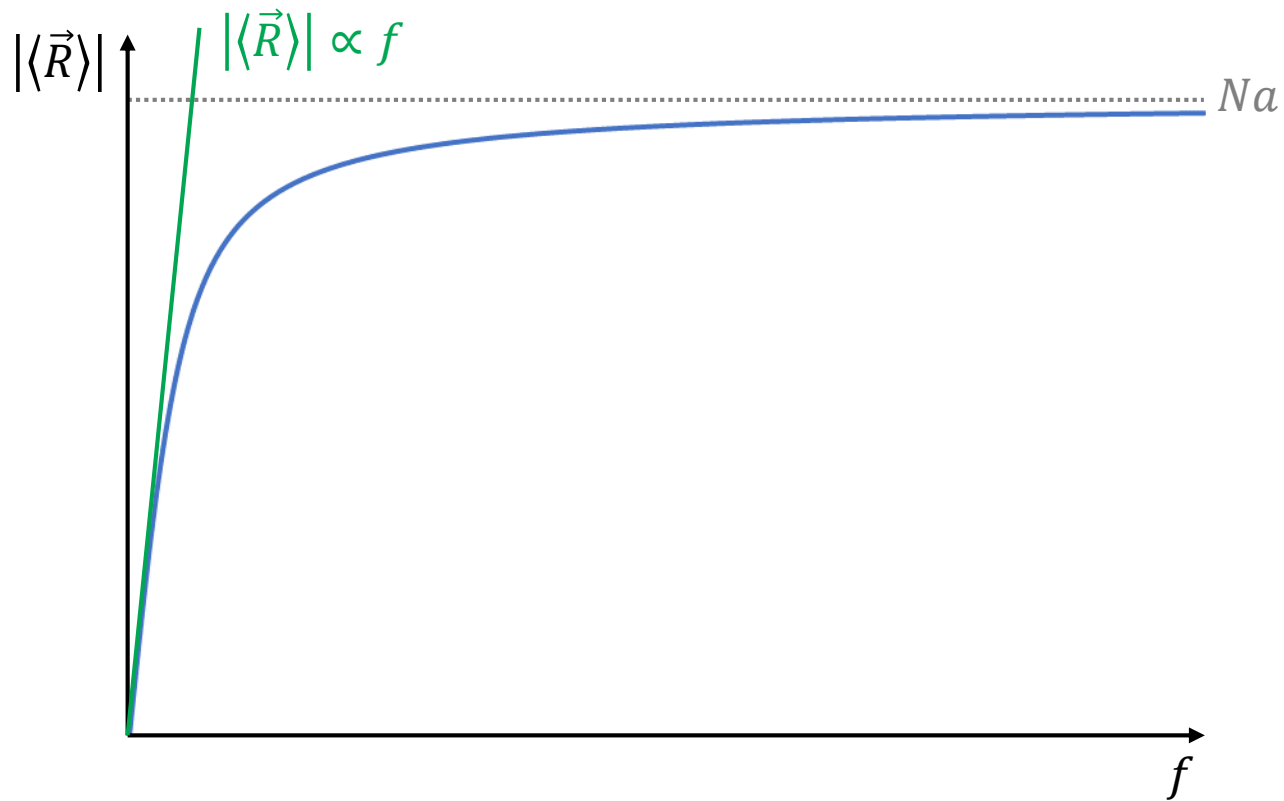
For small x , $\coth x \approx (1/x) + (x/3) + \dots$

Thus, at small f ,

$$|\langle \vec{R} \rangle| \approx Na \left(\frac{k_B T}{fa} + \frac{fa}{3k_B T} - \frac{k_B T}{fa} \right) = \frac{Na^2}{3k_B T} f$$

At large f , $\coth(fa/k_B T) \rightarrow 1$ and $|\langle \vec{R} \rangle| \rightarrow Na$

Force-Extension Curve



Worm-Like Chain (WLC)

(Kratky-Porod model)

- Freely jointed chain

$$E = -\vec{f} \cdot \sum_{i=1}^N \vec{r}_i$$

- Inclusion of neighbor interactions (cf. 1D Ising model)

$$E = -\vec{f} \cdot \sum_{i=1}^N \vec{r}_i - \kappa \sum_{\langle ij \rangle} \vec{r}_i \cdot \vec{r}_j$$

(κ : “stiffness”)

Difficult to solve!

End-to-end Distance for WLC

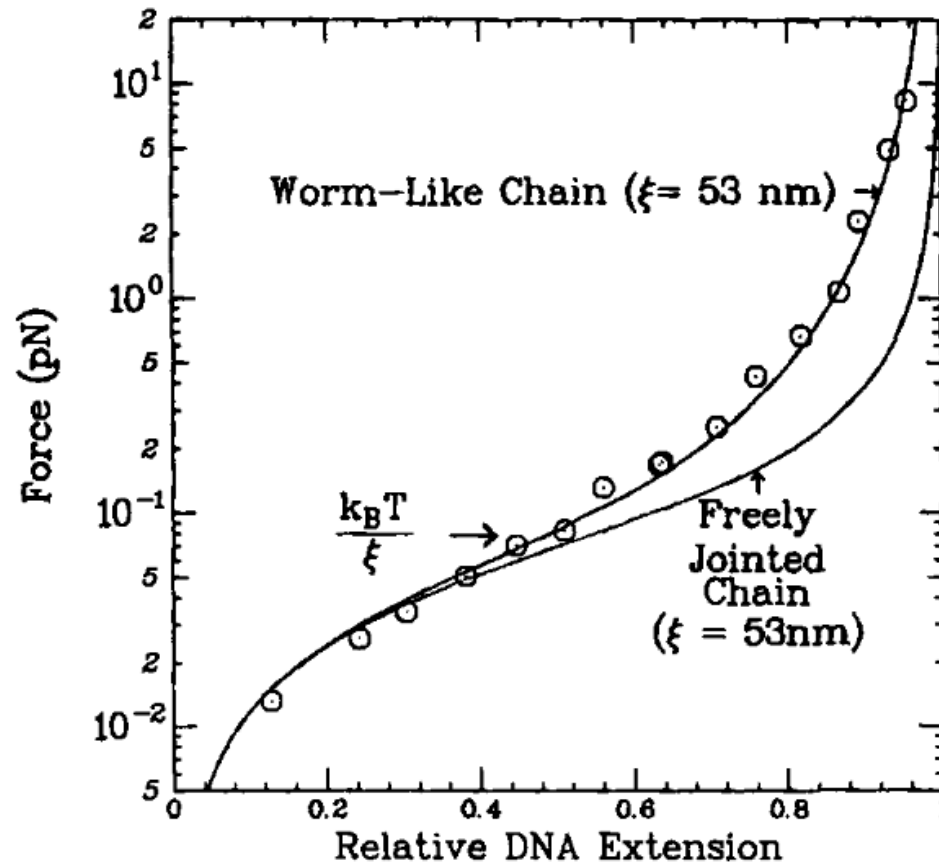
$$\langle \vec{R}^2 \rangle = \left\langle \sum_{i,j} \vec{r}_i \cdot \vec{r}_j \right\rangle = 2\xi [Na + (e^{-Na/\xi} - 1)\xi]$$

$$\xi = -\frac{a}{\ln[\coth(\beta\kappa a^2) - 1/(\beta\kappa a)]} \approx \frac{\kappa a^3}{k_B T}$$

$\uparrow$
 $\kappa a^2 \gg k_B T$

- $Na \gg \xi$: $\langle \vec{R}^2 \rangle \approx 2\xi Na \propto N$ (ideal chain)
- $Na \ll \xi$: $\langle \vec{R}^2 \rangle \approx (Na)^2 = R_{\max}^2$ (stiff rod)

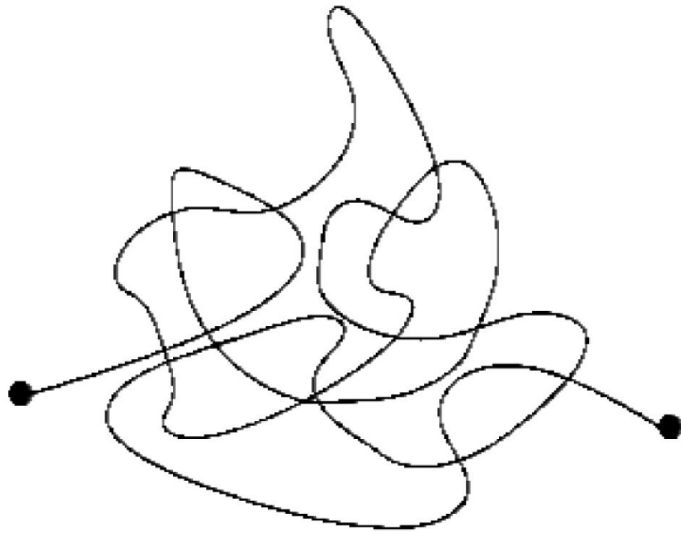
Force-Extension Curves of dsDNA



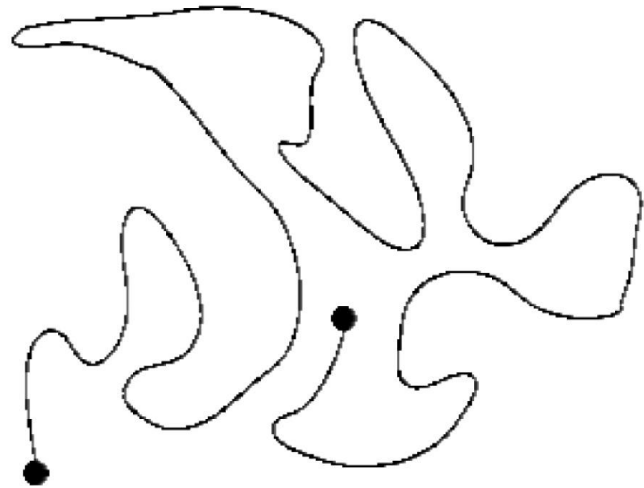
(Image: Strick et al., *J Stat Phys* (1998))

Long-Range Interactions

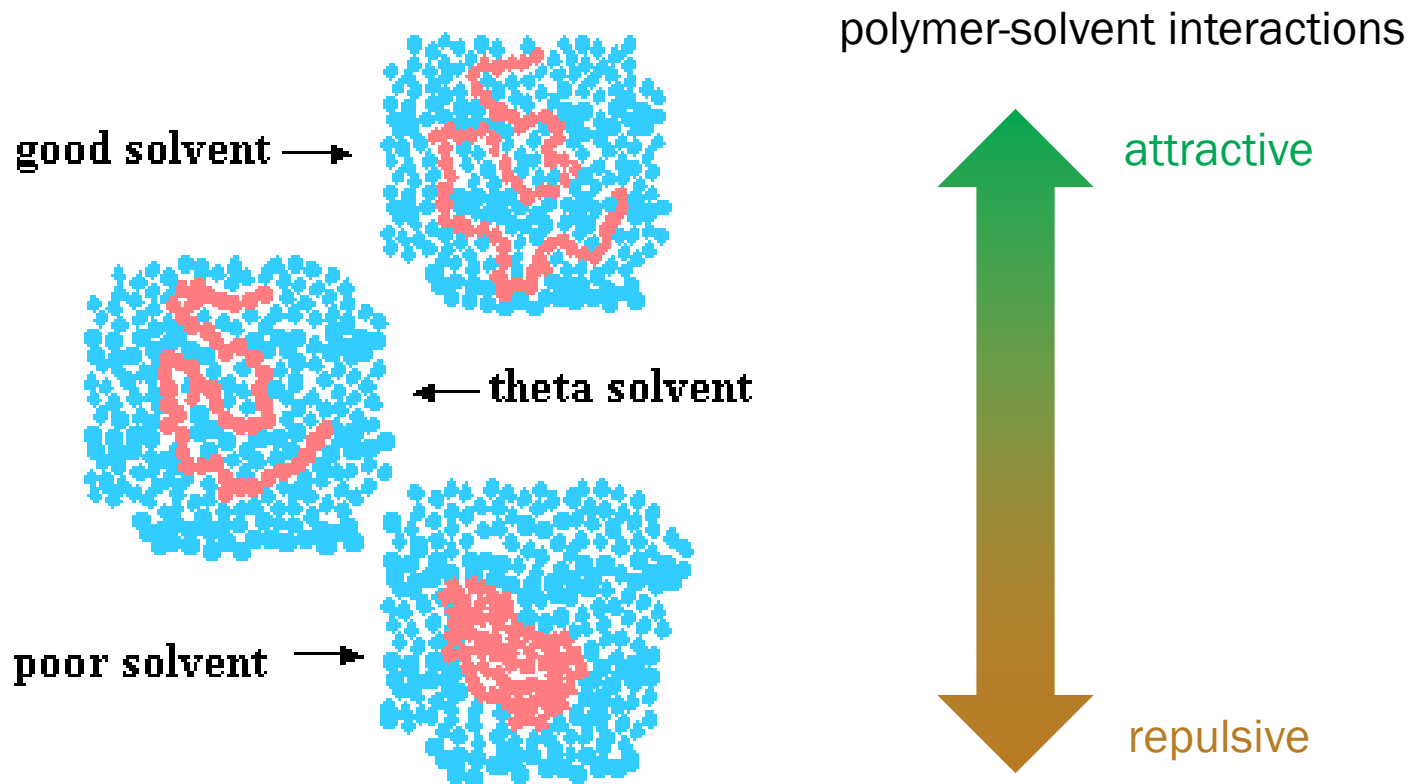
Ideal chains



“Real” chains



Origin of Long-Range Interactions



(Image: http://rkt.chem.ox.ac.uk/lectures/liqsolns/polymer_solutions.html)

Power Laws

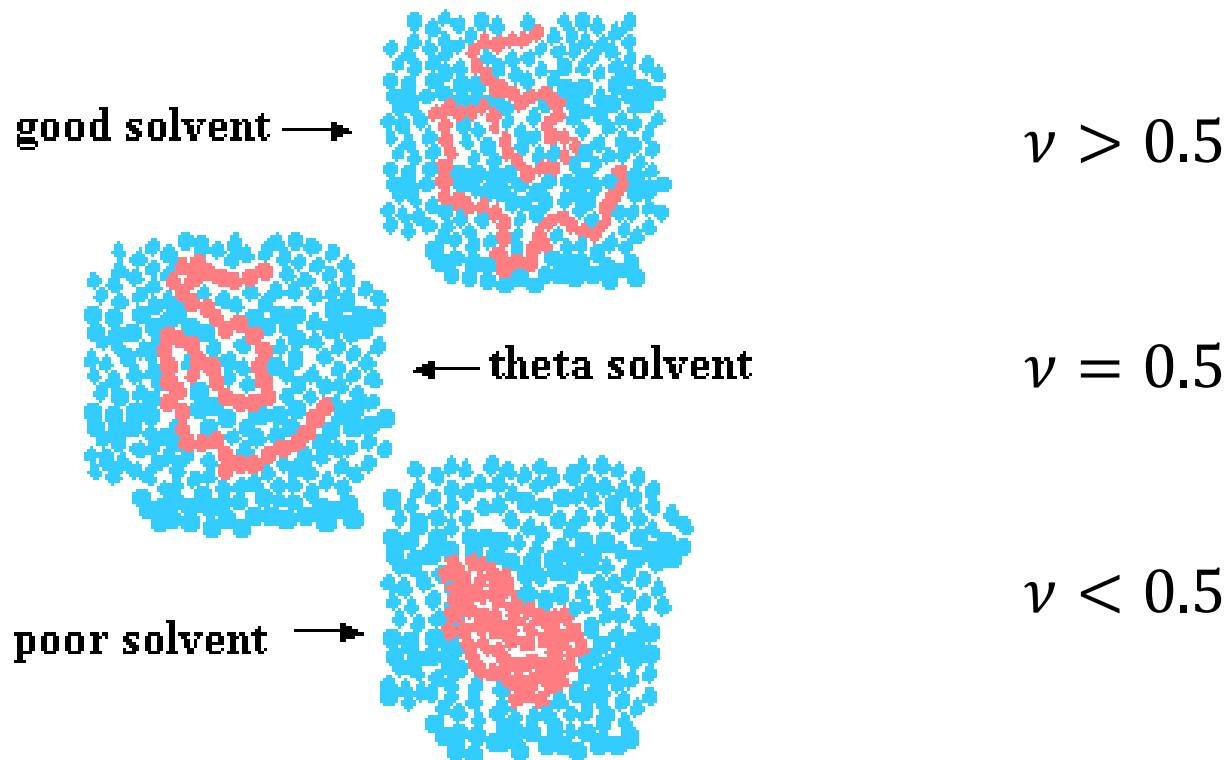
For a freely jointed chain,

$$\langle \vec{R}^2 \rangle^{1/2} = (Na^2)^{1/2} \propto N^{1/2}$$

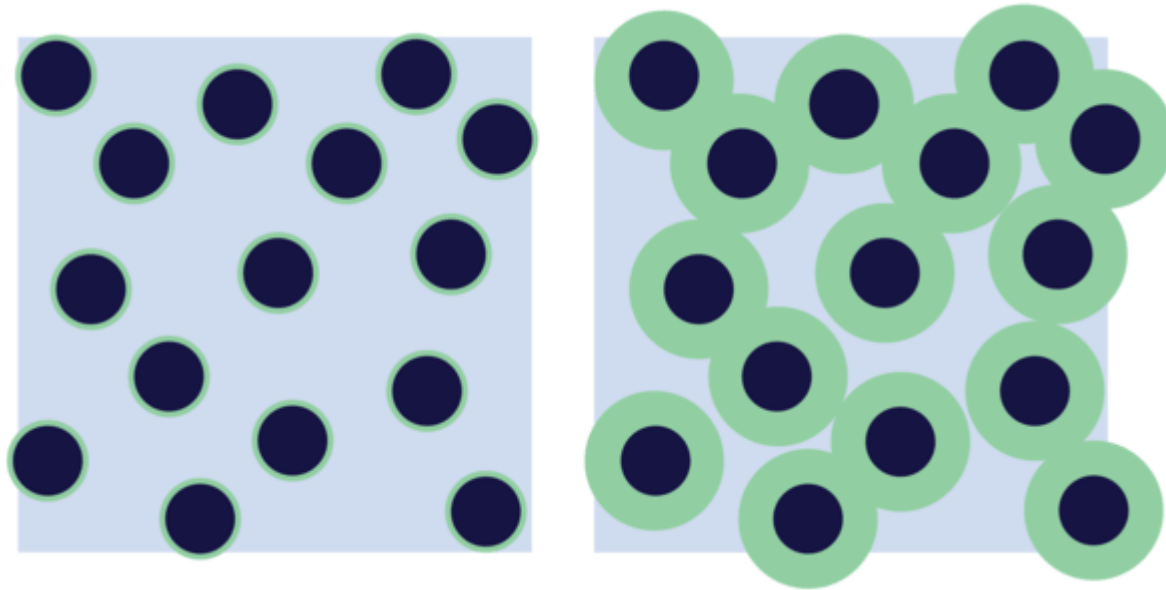
$$\langle R_g^2 \rangle^{1/2} = \left(\frac{Na^2}{6} \right)^{1/2} \propto N^{1/2}$$

Generally, $R \propto N^\nu$ (ν : Flory exponent)

Flory Exponents for Real Chains



Excluded Volume



Repulsion: positive excluded volume

Attraction: negative excluded volume

Flory Theory: Good Solvent

- Polymer with N monomers
- Each monomer occupies excluded volume of v_0
- Assume the polymer size $R \rightarrow$ volume $\sim R^3$
- Excluded-volume cost: $A_{\text{ev}} \approx k_B T \times v_0 (N/R^3) \times N$
- Entropic cost: $A_{\text{ent}} \approx k_B T R^2 / N a^2$ (blob argument)
- Free energy

$$A = A_{\text{ev}} + A_{\text{ent}} = k_B T \left(v_0 \frac{N^2}{R^3} + \frac{R^2}{N a^2} \right)$$

Minimization

$$A = A_{\text{ev}} + A_{\text{ent}} = k_{\text{B}}T \left(v_0 \frac{N^2}{R^3} + \frac{R^2}{Na^2} \right)$$

Minimizing free energy with respect to R ,

$$\frac{dA}{dR} = k_{\text{B}}T \left(-3v_0 \frac{N^2}{R^4} + 2 \frac{R}{Na^2} \right) = 0$$

$$R_{\text{good}} = v_0^{1/5} a^{2/5} N^{3/5} \propto N^{3/5}$$

(more accurate estimate: $\nu \approx 0.588$)

Flory Theory: Poor Solvent

$$A = A_{\text{ev}} + A_{\text{ent}} = k_{\text{B}}T \left(v_0 \frac{N^2}{R^3} + \frac{R^2}{Na^2} \right)$$

- Poor solvent: $v_0 < 0 \rightarrow$ minimization gives $R \approx 0$
- At the dense limit,

$$\frac{A_{\text{ev}}}{R^3} \approx k_{\text{B}}T(v_0 c^2 + w c^3 + \dots)$$

Considering the three-body interaction with $w > 0$,

$$A = k_{\text{B}}T \left(v_0 \frac{N^2}{R^3} + w \frac{N^3}{R^6} + \frac{R^2}{Na^2} \right)$$

Minimization

$$A = k_{\text{B}}T \left(v_0 \frac{N^2}{R^3} + w \frac{N^3}{R^6} + \frac{R^2}{Na^2} \right)$$

For $R \ll \sqrt{Na}$,

$$A \approx k_{\text{B}}T \left(v_0 \frac{N^2}{R^3} + w \frac{N^3}{R^6} \right)$$

Minimizing free energy with respect to R ,

$$R_{\text{poor}} = \left(\frac{wN}{|v_0|} \right)^{1/3} \propto N^{1/3}$$